

CV-Mora Based Lip Sync Facial Animations for Japanese Speech

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ABSTRACT

To generate authentic real-time facial animations using face mesh data, which corresponds to fifty-six consonant and vowel (CV) types of morae that form the basis of Japanese speech, we propose a new method. Our method produces facial expressions by the weighted addition of fifty-three face meshes based on the real-time mapping of voice streaming to registered morae. In the user study, results showed that facial expressions produced during Japanese speech were more natural using our method than those using popular methods to generate real-time English-based Oculus lip sync and volume intensity-based facial animations.

Index Terms: Computing methodologies—Computer graphics—Animation—Procedural animation; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

The metaverse era is fast approaching where many people can participate in a shared digital space. With the widespread of high-bandwidth next-generation network infrastructures, artificial intelligence avatars are now considered useful not only in virtual reality (VR) spaces but also in online e-commerce, customer service, teleteaching, and language training. In such cases, avatars must be equipped with the ability to speak with natural facial expressions to gain trust and empathy as human-like beings. A human-like natural avatar enhances the quality of social activities, including commercial services provided in virtual spaces.

Several studies have explored methods to synthesize the natural facial expressions of three-dimensional (3D) head models during speech, such as traditional linguistic analysis [2], more recent performance capture [6], and machine learning [4]. In these studies, focusing on the English language, the researchers could automatically create convincing facial animations corresponding to English speech sounds. However, they were not necessarily applied to speech animation in other languages that have different structures of pronunciation, especially in Japanese. This is because the number of phonemes and structure of the phoneme sequence within words and phrases in Japanese differ from English.

The Japanese language has a very restricted phonological inventory, which comprises five vowel phonemes (/a e i o u/) and sixteen consonant phonemes (/p t k b d g s h z r m n w j N Q/) [3]. Conversely, English comprises twenty-four consonants and many consonant clusters that sandwich any combination of twenty vowels. Japanese has a remarkable property: Japanese speech is perceived as a sequence of morae whose phoneme ends in a vowel, whereas English pronunciation is dominated more by consonants than vowels. In English, the stress structure of a syllable comprising a sequence of phonemes is important, whereas in Japanese, the mora as the

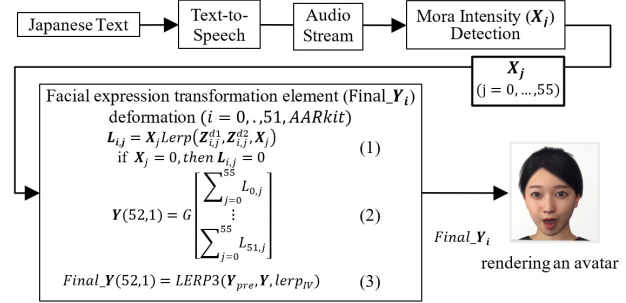


Figure 1: Procedure of speech facial animation using MDFE.

substructure of a syllable or a different segmentation level equivalent to a syllable is perceived as an articulation unit [5]. The mora corresponds to “kana” orthography as the basis of Japanese speech. A single kana (grapheme) is primarily pronounced as a pair of CV. Japanese speech is a series of sounds comprising kana or morae. The Japanese linguistic pattern is very different from that of English; therefore, facial shapes generated while speaking must be created differently than English.

2 LIP SYNC FACE GENERATION FOR A JAPANESE AVATAR

2.1 Mora-driven facial animation

The MDFE method was developed for an avatar that speaks Japanese. Figure 1 shows the process of MDFE. This speech animation system comprises a 3D neutral face mesh, fifty-two 3D face meshes for each facial expression, and text-to-speech with an audio processing system. The face animation while speaking is synthesized using the blendshape method that interpolates fifty-three 3D face meshes, including a neutral face mesh. The fifty-two 3D meshes were built using the Apple ARKit, using the ARFaceAnchor.BlendShapeLocation functionality. Each face mesh represents a particular expression, which is defined as a blend shape coefficient; fifty-two coefficients was set at the maximum value.

The set of the deformation coefficients was obtained from the distance sensor (Apple TrueDepth camera system) of the iPad Pro and stored as a vector $\mathbf{AAR}(52,1)$ after it was sent to a PC via the user datagram protocol. The CV-mora facial expression matrix $\mathbf{Y}(52,56)$ was built based on the facial measurements during each of the fifty-six morae utterances performed by a single assistant of the experiment. The pronounced sound of a single Japanese kana grapheme that corresponds to one mora was recorded, and its acoustic feature was calculated and stored in twelve-dimensional mel-frequency cepstrum coefficients (MFCCs) at 100 Hz. First, we registered the average dimension of the MFCC for each of the fifty-six morae for 1024 ms during kana (mora) pronunciation. The cosine similarity of the input audio to profile p determines the mora intensity vector $\mathbf{X}_j (0 \leq j \leq 55, 0 \leq \mathbf{X}_j \leq 1.0)$ at 100 Hz.

Additionally, the face mesh for each mora was predetermined and stored in the CV-mora expression matrix $\mathbf{Z}(52,56)$ that is used as a part of the facial expression in real time. The matrix $\mathbf{Z}$ comprises the fifty-six sets of facial deformation vector $\mathbf{AAR}$ for every mora. The fifty-six $\mathbf{AAR}$ s are differences from the neutral face mesh during

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Japanese mora pronunciation. The shape of the face starts to deform at 120 ms before vocalization, and the deformation reaches the apex at the onset of vocalization [1]. To incorporate this process, we captured two AARs at 120 ms before vocalization onset and at the apex and obtained $Z_{i,j}^{d1}$ and $Z_{i,j}^{d2}$ ($0 \leq i \leq 51$) for each mora j ($0 \leq j \leq 55$), respectively. The apex defines the absolute sum of the elements in **AAR**, which reaches the maximum within the utterance of the mora. The vectors $Z_{i,j}^{d1}$ and $Z_{i,j}^{d2}$ for each mora j were added using a linear interpolation function (*Lerp*) with mora intensity X_j as the weight (1), as shown in Figure 1. Vector Y denotes the facial transformation process to create a face mesh to render as determined using Equation (2), which calculates the weighted elements from every mora. The result passes through a cubic interpolation function (3) with previous vector Y_{pre} to avoid unnaturalness caused by high-frequency deformation.

2.2 Three other methods for comparison

We used three other methods to compare with our method in the user study. A method that simplifies MDFE is only based on the five vowels of Japanese: vowel-driven facial expression (VDFE). The five vowels are especially important for Japanese language because any of the five vowels is involved in every kana grapheme from which Japanese speech is composed. The VDFE system detects the five vowel components and produces a face mesh by the weighted sum of the five face meshes for the vowels. To determine the optimal method, a face mesh with the jaw opened proportional to the magnitude of vocalization obtained from Audio-filter was also implemented: volume method. Additionally, the Oculus lip sync model, a well-known real-time speech expression synthesis method, was used: Oculus method.

3 USER STUDY AND RESULTS

We evaluated our CV-mora based lip sync facial animation system (MDFE) in comparison with the three other methods: volume, Oculus lip sync, and VDFE. The items of evaluation were the accuracy of mouth shape and naturalness (human likeness) of the face while speaking. The participants of this experiment were fifteen Japanese students and faculty members (twelve males, three females) with an average age of 24.1 y.

The stimuli were four facial animations in a random order with the sound of Japanese speech for 21 s. The visual presentation was performed in a VR space using a head-mounted display. The monaural voice was output through head phones.

The evaluation results was written on a visual analogue scale (VAS) that was anchored (1) “completely mismatch” to “completely match” for accuracy, (2) “robot-like and expressionless” to “natural expression of human speech” for naturalness from left to right on the scale, respectively. Additionally, the participants were asked to rank the four methods regarding these two items.

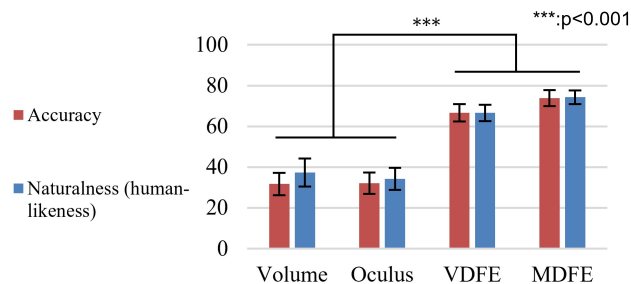


Figure 2: Accuracy of mouth shape and naturalness of facial expression during Japanese speech (N=15).

Figure 2 shows the results. The two-way analysis of variance model revealed that the main effect of the animation method was highly significant ($p < 0.001$), whereas the evaluation item was not significant. The multiple comparison (Holm’s method) indicated highly significant difference between the two groups of MDFE/VDFE and volume/Oculus lip sync ($p < 0.001$). The ranking results indicated that the order was MDFE, VDFE, volume, and Oculus from the first to the last according to the average ranking score.

4 DISCUSSION

MDFE, based on fifty-six CV-type morae, was ranked as the highest in the order evaluation. VDFE ranked second, which is exceedingly better than the volume/Oculus group according to the accuracy and naturalness evaluation (Figure 2). The fidelity of speech expression in VDFE was less than MDFE, which uses face shape data for the five morae. VDFE often created the same facial motion with an insufficient number of facial expression variations. This result may have caused robot-like unnatural impression when compared to MDFE. However, there was neither a significant difference between MDFE/VDFE in the rank test nor in the multiple comparisons of VAS, which may suggest that the most participants may have overlooked small differences under the limited duration of observation.

Interestingly, the Oculus lip sync and volume methods had almost the same low-level results. The Oculus method detects the English phonemes that are not necessarily applied to Japanese morae. When considering the volume method, which simply opens the jaw to the magnitude of the voice, the Oculus and volume methods are not in the same direction even if they are both far from the correct facial expressions.

5 FUTURE WORK

We demonstrated that mora-based facial animation is suitable for Japanese speech. The characteristics of MDFE components must be investigated to improve the avatar’s facial expression to be closer to the natural human face in terms of speech styles. Moreover, the effectiveness of this method will be demonstrated through comparisons with previous studies.

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